

MODULE 1

CYBERCRIME: THE PHENOMENON

AI and ML and cyber-crime

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Human-machine interaction and crime

The literature on digital sociology emphasizes the **relationship between the human being and the machine in the construction of the new digital society**, in a socio-technical vision. In a world where robots support or replace us, in which artificial intelligence is gaining ground, one therefore wonders if there are or will be more and more hybrid criminals, the result of the close interaction between humans and computers. We are referring, for example, to phenomena, as King et al. define them (2020), of "**crime perpetrated by artificial intelligence**" in which the perpetrator is very difficult to circumscribe, with his characteristics, and in which some questions arise spontaneously: "**who commits the crime [...]?** A human agent? An artificial agent? Both of them?" (ibid.: 91).

Artificial intelligence: the concept

Artificial intelligence deals with **artificial agents capable of carrying out tasks that usually require intelligence and human intervention to be conducted**. These artificial agents are “**sufficiently informed**”, **intelligent**”, “autonomous and capable of carrying out morally relevant actions independently of the humans who created them” (Floridi and Sanders, 2004: 351).

Artificial intelligence: new challenges for crime and criminology

- Crime with artificial intelligence
- Crime against artificial intelligence
- Crime through artificial intelligence

Crime with artificial intelligence

As Brundage et al. (2018: 19-22) artificial intelligence in this case can be a powerful tool for criminal purposes, because **it is capable of expanding existing threats**, introducing entirely **new threats** or **completely altering the characteristics of threats**.

Example: **deepfake**.

<https://www.youtube.com/watch?v=CDMVaQOvtxU> (Tom Cruise)

<https://www.youtube.com/watch?v=cQ54GDm1eL0> (President Obama)

<https://www.youtube.com/watch?v=gLoI9hAX9dw> (various examples)

Crime against artificial intelligence

In this case we are talking about **attacks that exploit the vulnerabilities of an artificial intelligence system to make fun of or “hypnotize” the system itself**. Thus, for example, it is possible to “poison” the training data that are provided to the machine to train it and program the artificial intelligence to commit criminal acts, behaving as an autonomous criminal agent.

Example: Microsoft’s Tay.

<https://www.youtube.com/watch?v=HsLup7yy-6I>

Crime through artificial intelligence

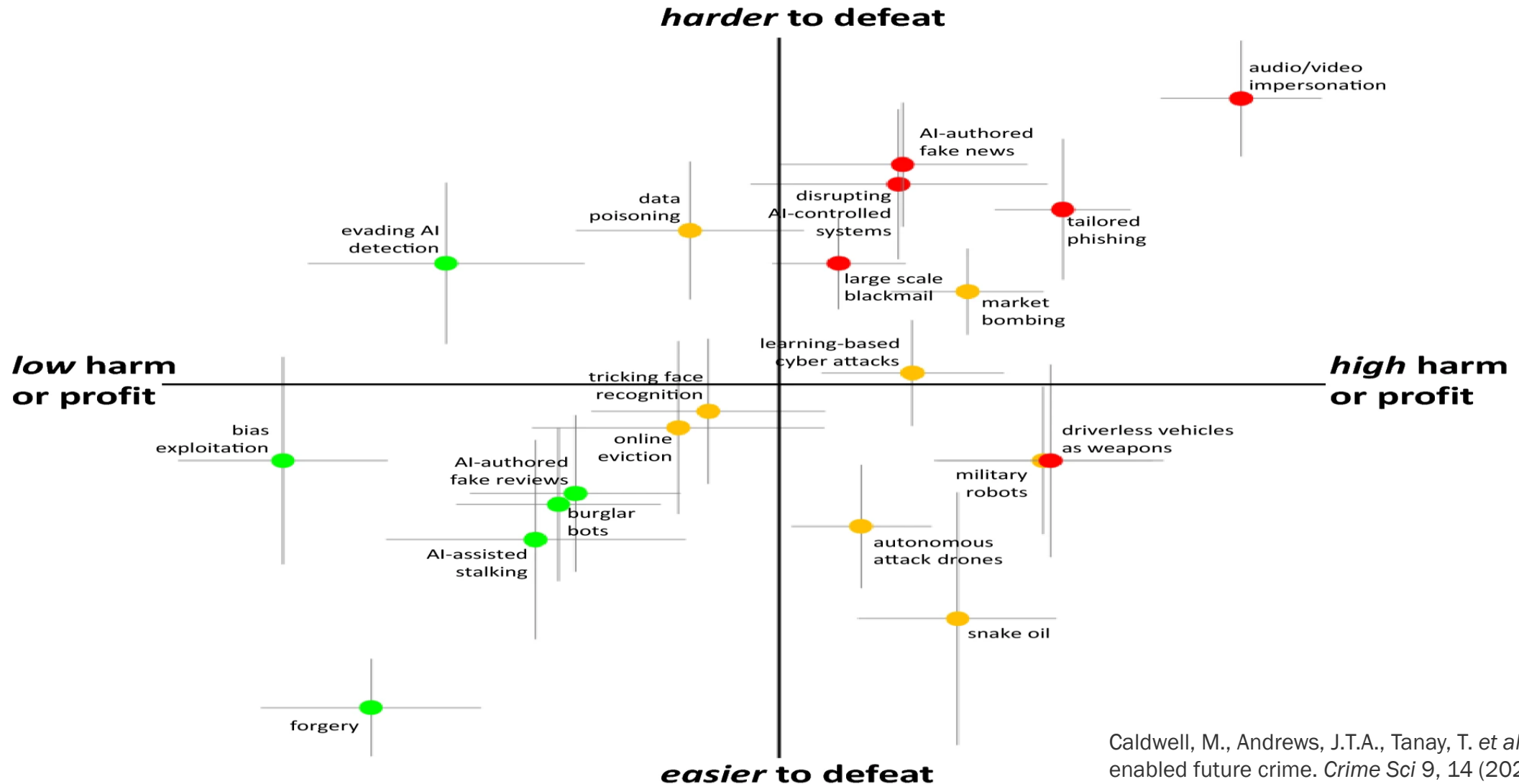
In this case AI becomes an **intermediary agent**.

For example, a **bot can buy drugs on the dark web** instead of a natural person. That is, artificial intelligence can act as a screen, a criminal shield.

AI and crime areas

- Identity theft and fraud, impersonation and more generally online fraud
- Financial crime sector
- Drug trafficking and trading
- Harassment
- Sex crimes

AI-enabled future crimes: high, medium, low risk



Caldwell, M., Andrews, J.T.A., Tanay, T. et al. AI-enabled future crime. *Crime Sci* 9, 14 (2020).

Cyborg-crime: the concept

Working on botnets (networks of infected computers controlled by a controller that are increasingly used to commit cyber-crime) van der Wagen and Pieters (2015) coined the concept of “cyborg-crime”, that is, online crime perpetrated by a **cyborg actor** and, above all, they proposed using the **Actor-Network Theory** as a perspective to interpret these **forms of delinquency committed by actors who are neither completely human, nor completely driven by a machine.**

The two authors argue that **it is not possible to understand the nature of the crimes carried out through botnets if we continue to use an anthropocentric criminology** that sees the human being as the main force behind criminal behavior. “Rather than considering a botnet as a technological tool, an individual (opportunistic) crime or a resource on the criminal market, **we treat a botnet as a hybrid crime network, a crime that derives from mutual cooperation and human / technology interaction**” (ibid. : 579).

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